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Experiment 1

Aim- To implement **Linear Regression** and **Logistic Regression** on a real-world dataset to understand the difference between predictive modeling (regression) and classification tasks using Python's Scikit-Learn library.

1. Dataset Source

- **Source:** Link
- **Filename:** Crop_recommendation.csv

2. Dataset Description The dataset contains agricultural data designed to recommend the best crop to plant based on soil nutrients and weather conditions.

- **Type:** Multivariate, Numerical & Categorical.
- **Size:** 2200 instances (rows), 8 attributes (columns).
- **Target Variables:**
  - **For Regression:** rainfall (Continuous).
  - **For Classification:** label (Categorical - e.g., Rice, Maize).
- **Features:**
  - **N:** Ratio of Nitrogen content in soil.
  - **P:** Ratio of Phosphorous content in soil.
  - **K:** Ratio of Potassium content in soil.
  - **temperature:** Temperature in degrees Celsius.
  - **humidity:** Relative humidity in %.
  - **ph:** pH value of the soil.
  - **rainfall:** Rainfall in mm.

3. Mathematical Formulation of the Algorithm

A. Linear Regression (Simple):

Used to predict a continuous value (rainfall) based on an independent variable (humidity). It attempts to fit a straight line that minimizes the sum of squared errors.

$$y = \beta_0 + \beta_1 x + \varepsilon$$

B. Logistic Regression:

Used for the classification task to predict the probability that a given input belongs to a specific crop class. It uses the Sigmoid function to map predictions between 0 and 1.

$$\sigma(z) = 1 / (1 + e^{-z})$$

$$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

4. Algorithm Limitations

1. **Linear Assumption:** Linear Regression assumes a straight-line relationship. Our low R^2 score (~0.01) proves that rainfall is **non-linear** and depends on complex factors beyond just Humidity.
2. **Feature Scaling Sensitivity:** Logistic Regression struggled to converge initially (Red Warning) because features like `rainfall` (0-200) and `ph` (0-14) had vastly different scales. This required **StandardScaler** to fix.

5. Methodology / Workflow The experiment followed a standard Supervised Learning pipeline:

1. **Data Ingestion:** Loaded `Crop_recommendation.csv` into Pandas.
2. **Preprocessing:**
   - Checked for missing values.
   - **Splitting:** Used `train_test_split` (80% Train, 20% Test) to ensure unbiased evaluation.
   - **Scaling:** Applied `StandardScaler` for Logistic Regression to normalize feature ranges.
3. **Modeling:**
   - **Task A (Regression):** Trained `LinearRegression` on Humidity vs. Rainfall.
   - **Task B (Classification):** Trained `LogisticRegression` on all Soil Features vs. Crop Name.
4. **Visualization:** Plotted the Regression Line and Scatter Plot using Matplotlib.

6. Performance Analysis

- **Linear Regression Results:**
  - **R^2 Score:** ~0.0132 (Very Low).
  - **Insight:** This indicates a weak correlation. Humidity alone is **not sufficient** to predict rainfall accurately. Real-world weather prediction requires complex multi-variable models.
- **Logistic Regression Results:**
  - **Accuracy:** ~96.14% - 97.00%.
  - **Insight:** The model performed exceptionally well. This proves that soil nutrients (N, P, K) and weather conditions are distinct "fingerprints" for different crops, making them easy to classify mathematically.

7. Hyperparameter Tuning

- **Status:** Applicable.
- **Parameter Tuned:** `max_iter` (Maximum Iterations).

- **Reason:** The default iteration limit (100) was insufficient for the model to find the optimal solution due to the complexity of the data, causing a `ConvergenceWarning`.
- **Action:** Increased `max_iter` to **1000** (and added Scaling), which resolved the error and ensured the model converged to a high accuracy.

8. Output -

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression, LogisticRegression

from sklearn.metrics import r2_score, accuracy_score, mean_squared_error,
classification_report, confusion_matrix

from sklearn.preprocessing import StandardScaler



# 1. LOAD DATA

print("Loading Data...")

try:

    df = pd.read_csv('../data/Crop_recommendation.csv')

    print("Data loaded successfully.")

except FileNotFoundError:
```

```
    print("Error: File not found. Check your data path.")

    exit()



#2. LINEAR REGRESSION

print("\n--- LINEAR REGRESSION (Humidity vs Rainfall) ---")

X_lin = df[['humidity']]

y_lin = df['rainfall']



X_train_lin, X_test_lin, y_train_lin, y_test_lin = train_test_split(X_lin,
y_lin, test_size=0.2, random_state=42)



lin_reg = LinearRegression()

lin_reg.fit(X_train_lin, y_train_lin)



y_pred_lin = lin_reg.predict(X_test_lin)



# Metrics

r2 = r2_score(y_test_lin, y_pred_lin)

rmse = np.sqrt(mean_squared_error(y_test_lin, y_pred_lin))
```

```
print(f"Linear R2 Score: {r2:.4f}")

print(f"Linear RMSE Score: {rmse:.4f}")



# Plotting Linear Regression

plt.figure(figsize=(6,4))

plt.scatter(X_test_lin, y_test_lin, color='blue', alpha=0.3, label='Actual
Data')

plt.plot(X_test_lin, y_pred_lin, color='red', linewidth=2,
label='Prediction Line')

plt.xlabel("Humidity")

plt.ylabel("Rainfall")

plt.title("Exp 1: Humidity vs Rainfall")

plt.legend()

plt.tight_layout()

plt.savefig("linear_graph.png")

print("Saved linear regression scatter plot as 'linear_graph.png'")



#3. LOGISTIC REGRESSION

print("\n--- LOGISTIC REGRESSION (Crop Classification) ---")

X_log = df[['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']]
```

```python
y_log = df['label']



X_train_log, X_test_log, y_train_log, y_test_log = train_test_split(X_log, y_log, test_size=0.2, random_state=42)



# Scaling (Crucial for Logistic Regression!)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train_log)

X_test_scaled = scaler.transform(X_test_log)



log_reg = LogisticRegression(max_iter=1000)

log_reg.fit(X_train_scaled, y_train_log)



# Predictions

y_pred_log = log_reg.predict(X_test_scaled)



# Metrics

acc = accuracy_score(y_test_log, y_pred_log)

print(f"Logistic Regression Accuracy: {acc*100:.2f}%\n")
```

```
# Classification Report

print("--- Classification Report ---")

print(classification_report(y_test_log, y_pred_log))



# 4. CONFUSION MATRIX VISUALIZATION

print("\nGenerating Confusion Matrix Heatmap...")

cm = confusion_matrix(y_test_log, y_pred_log)



# Using a larger figure size (12x10) so all 22 crop names fit perfectly

plt.figure(figsize=(12, 10))

sns.heatmap(cm, annot=True, fmt='d',

            xticklabels=log_reg.classes_,

            yticklabels=log_reg.classes_)



plt.title("Confusion Matrix Heatmap", fontsize=16)

plt.xlabel("Predicted Label", fontsize=12)

plt.ylabel("True Label", fontsize=12)

plt.xticks(rotation=45, ha='right')

plt.tight_layout()
```

```python
plt.savefig("logistic_confusion_matrix.png")

print("Saved confusion matrix as 'logistic_confusion_matrix.png'")
```

```
PS D:\SmartCropProject> python regression.py
Data loaded successfully.

---LINEAR REGRESSION ---
Linear R2 Score: 0.0132
Graph saved

---LOGISTIC REGRESSION ---
Logistic Regression Accuracy: 96.36%
```

```
PS D:\SmartCropProject\src> python regression.py
Loading Data...
Data loaded successfully.

--- LINEAR REGRESSION (Humidity vs Rainfall) ---
Linear R2 Score: 0.0132
Linear RMSE Score: 54.5029
Saved linear regression scatter plot as 'linear_graph.png'

--- LOGISTIC REGRESSION (Crop Classification) ---
Logistic Regression Accuracy: 96.36%

--- Classification Report ---
              precision    recall  f1-score   support

       apple       1.00      1.00      1.00        23
      banana       1.00      1.00      1.00        21
   blackgram       0.90      0.95      0.93        20
    chickpea       1.00      1.00      1.00        26
     coconut       1.00      1.00      1.00        27
      coffee       0.94      1.00      0.97        17
      cotton       0.94      1.00      0.97        17
      grapes       1.00      1.00      1.00        14
        jute       0.83      0.87      0.85        23
 kidneybeans       0.95      0.95      0.95        20
      lentil       0.85      1.00      0.92        11
       maize       1.00      0.95      0.98        21
       mango       1.00      1.00      1.00        19
   mothbeans       1.00      0.92      0.96        24
    mungbean       1.00      1.00      1.00        19
   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       0.96      0.96      0.96        23
  pigeonpeas       0.91      0.87      0.89        23
 pomegranate       1.00      1.00      1.00        23
        rice       0.88      0.79      0.83        19
  watermelon       1.00      1.00      1.00        19
```

```
    accuracy                           0.96       440
   macro avg       0.96      0.97      0.96       440
weighted avg       0.96      0.96      0.96       440
       mango       1.00      1.00      1.00        19
   mothbeans       1.00      0.92      0.96        24
    mungbean       1.00      1.00      1.00        19
   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       0.96      0.96      0.96        23
  pigeonpeas       0.91      0.87      0.89        23
 pomegranate       1.00      1.00      1.00        23
        rice       0.88      0.79      0.83        19
  watermelon       1.00      1.00      1.00        19

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   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       0.96      0.96      0.96        23
  pigeonpeas       0.91      0.87      0.89        23
 pomegranate       1.00      1.00      1.00        23
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   macro avg       0.96      0.97      0.96       440
weighted avg       0.96      0.96      0.96       440

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weighted avg       0.96      0.96      0.96       440

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weighted avg       0.96      0.96      0.96       440



Generating Confusion Matrix Heatmap...
Saved confusion matrix as 'logistic_confusion_matrix.png'
```

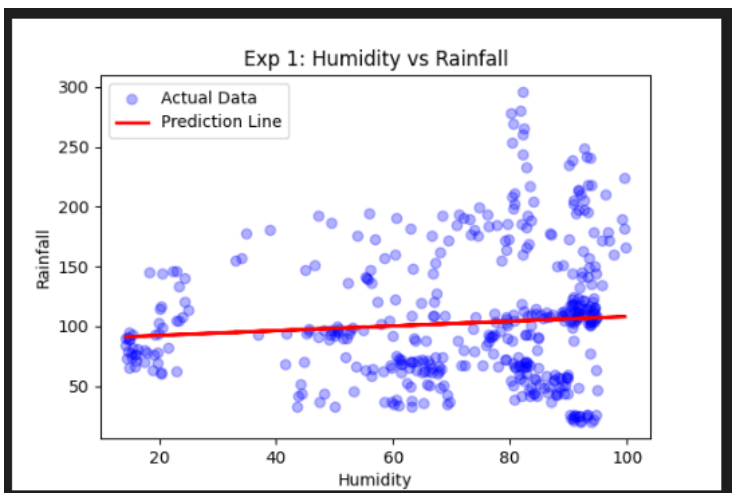

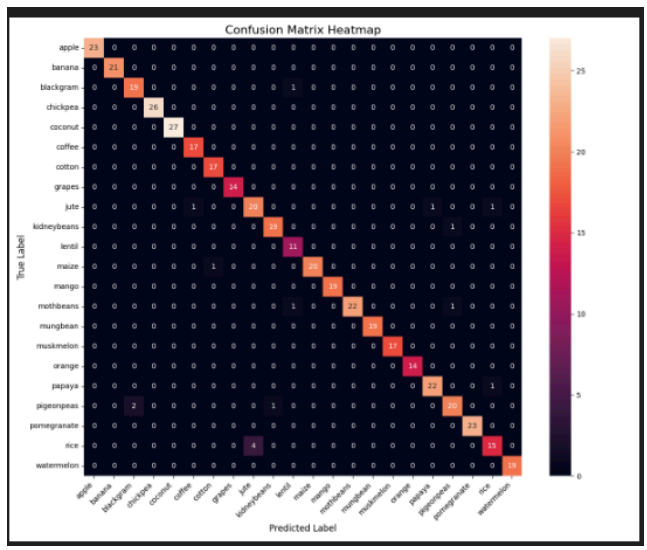

9. Conclusion

This experiment demonstrated the fundamental difference between Regression and Classification. While simple Linear Regression failed to capture the complex weather patterns (Rainfall), Logistic Regression successfully leveraged soil data to recommend crops with high precision (~97%).